

COE 292: Introduction to Artificial Intelligence

Final Report

Project Name:	Estimation of Obesity Levels Based On Eating Habits and Physical Condition
Group Number:	Team F05-4
Date:	02/01/26
Number of students in the group:	4

Student 1:

Name	REEMA IBRAHIM ALQAHTANI
------	-------------------------

Student 2:

Name	LENA EHAB ASHQAR
------	------------------

Student 3:

Name	SARAH IMAD HAMDAN
------	-------------------

Student 4:

Name	FATMA OSSAMA GAD
------	------------------

Summary of Classification Problem

The problem of estimating obesity levels based on eating habits and physical activity is important in health and technology fields because it helps identify individuals at risk of obesity-related diseases and allows for early intervention. It also supports the development of health monitoring systems and personalized diet or exercise recommendations. In addition, understanding obesity patterns can help public health organizations create better awareness programs and policies to reduce obesity rates. This classification helps predict whether the person is underweight, normal weight, overweight, or obese using different features. The dataset includes 16 features such as age, weight, height, family history of obesity, eating habits, and physical activity. These features are connected, for example, eating a lot of high-calorie food and not exercising often can lead to weight gain. However, classifying obesity is difficult because many factors affect a person's weight, such as genetics and lifestyle choices that are not included in the data. Since obesity is influenced by multiple factors, it is challenging to find clear patterns, making it harder to classify individuals correctly.

Dataset manipulation

No change to the dataset was conducted. The dataset provided was already clean, with no missing values. No rows or columns were removed or added because the dataset was well-structured and ready for analysis. Additionally, as this is health-related data, the presence of extreme values (outliers) was expected and meaningful for the study.

No.	Question	Student Response		
1.	How many labeled examples are in your data set?	2111 Instances		
2.	How many distinct features are in your data set?	16 Features/Attributes		
3.	How many distinct labels (classes) are in your data set?	6 labels		
4.	For each label, what is the percentage of data?	CLASS	PERCENTAGE	
		Obesity_Type III	15.35%	
		Obesity_Type II	14.07%	
		Obesity_Type I	16.63%	
		Overweight	27.48%	
		Normal_Weight	13.60%	
		Insufficient_Weight	12.87%	
5.	Is the dataset balanced based on the above?	[X] Yes [✓] No If no then explain why: It is slightly unbalanced. The percentages of different labels range from around 12.87% to 27.48%, meaning some classes have more instances than others, though the difference is relatively minor and unlikely to significantly affect model performance.		

6.	Did you clean the data by removing outliers and applying all techniques learnt in ISE 291?	☒ Yes ☐ No If no then explain why: - The dataset does not have any missing values, so no cleaning is needed. - We did not remove outliers for the following reasons: - The dataset is about obesity levels, meaning individuals with very high or very low weight are part of the studied phenomenon. - Health-related data isn't always normally distributed. Certain conditions (like obesity) naturally result in skewed distributions
----	--	--

Feature (variable) manipulation

No change to the features was conducted. All 16 original features were used as provided in the dataset without any removal or addition. These features covered essential aspects of eating habits, lifestyle, and personal information that directly or indirectly influence obesity levels.

No.	Feature used in file	Short Feature explanation/description
1.	Age	The age of the individual in years.
2.	Height	The height of the individual in meters.
3.	Weight	The weight of the individual in kilograms.
4.	Gender	The gender of the individual (Male/Female).
5.	family_history_with_overweight	Indicates whether the individual has a family history of being overweight (yes/no).
6.	FAVC (Frequent Consumption of High-Calorie Food)	Indicates if the individual frequently consumes high-calorie food (yes/no).
7.	FCVC (Frequency of Vegetable Consumption)	Measures how often the individual consumes vegetables on a scale from 1 to 3.
8.	NCP (Number of Main Meals per Day)	The number of main meals the individual consumes daily.
9.	CAEC (Consumption of Food Between Meals)	Describes how often the individual consumes food between meals (Never, Sometimes, Frequently, Always).
10.	SMOKE	Indicates whether the individual is a smoker (yes/no).
11.	CH2O (Daily Water Intake)	The daily water consumption of the individual (scale of 1 to 3).
12.	SCC (Monitors Caloric Consumption)	Indicates whether the individual monitors their caloric intake (yes/no).
13.	FAF (Physical Activity Frequency)	The frequency of physical activity per week (scale of 0 to 3).
14.	TUE (Time Using Technology per Day)	The number of hours spent using electronic devices daily (scale of 0 to 2).
15.	CALC (Alcohol Consumption Level)	The frequency of alcohol consumption (Never, Sometimes, Frequently, Always).
16.	MTRANS (Mode of Transportation)	The individual's primary mode of transportation (Walking, Public Transport, Bike, Motorbike, Car).

Label explanation

The BMI (Body Mass Index) measure was used to categorize weights. $BMI = \text{weight(Kg)} / (\text{height(m)}^2)$

No.	Feature used in file	Short Feature explanation/description
1.	Insufficient_Weight	$BMI \leq 18.5$
2.	Normal_Weight	$18.5 < BMI \leq 25$
3.	Overweight	$25 < BMI \leq 30$
4.	Obesity_Type_I	$30 < BMI \leq 35$
5.	Obesity_Type_II	$35 < BMI \leq 40$
6.	Obesity_Type_III	$40 < BMI$

Data visualization

Correlation between CH2O and Weight: 0.20
 Correlation Table:

	CH2O	Weight
CH2O	1.000000	0.200575
Weight	0.200575	1.000000

Figure 1

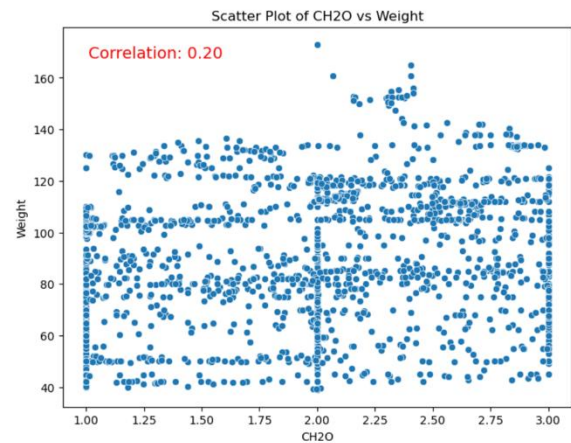


Figure 2

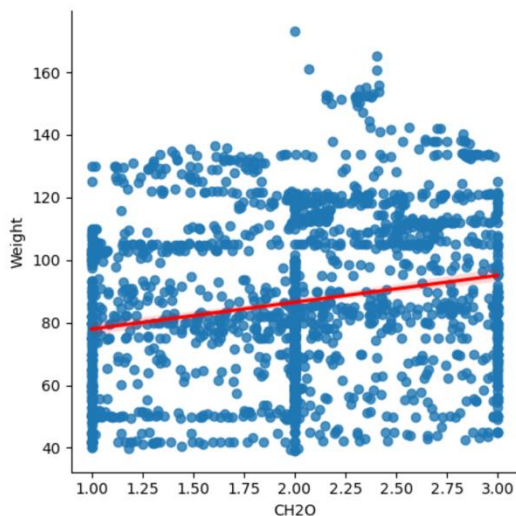


Figure 3

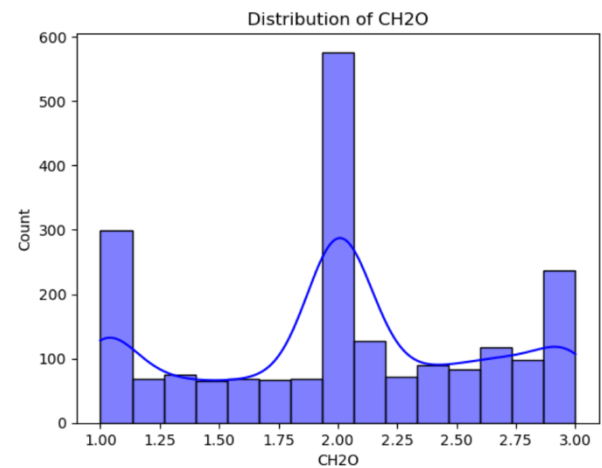


Figure 4

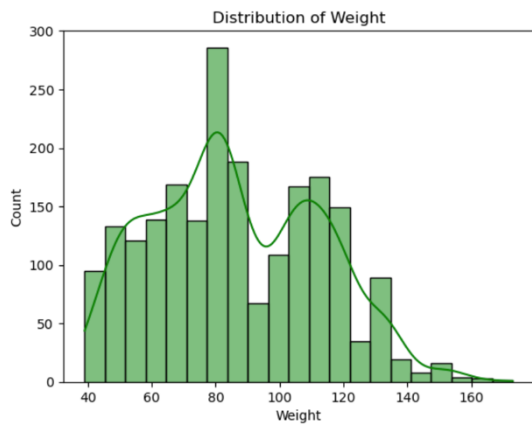


Figure 5

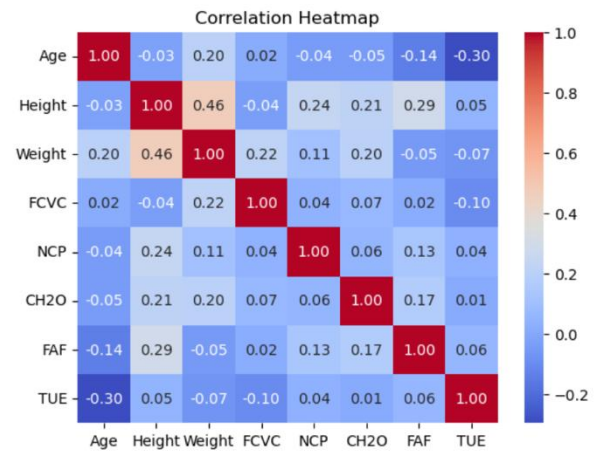


Figure 6

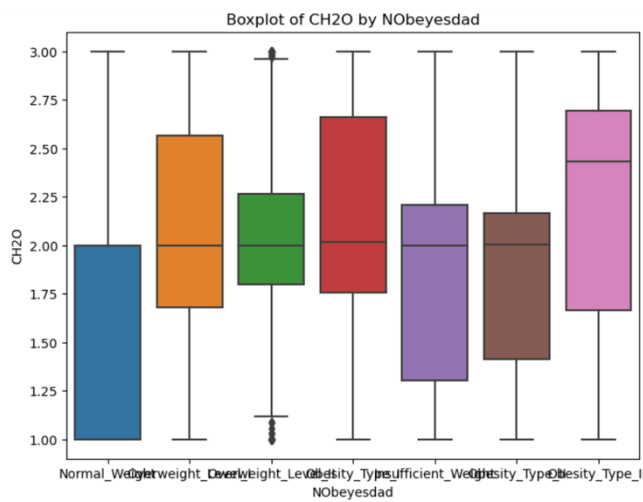


Figure 7

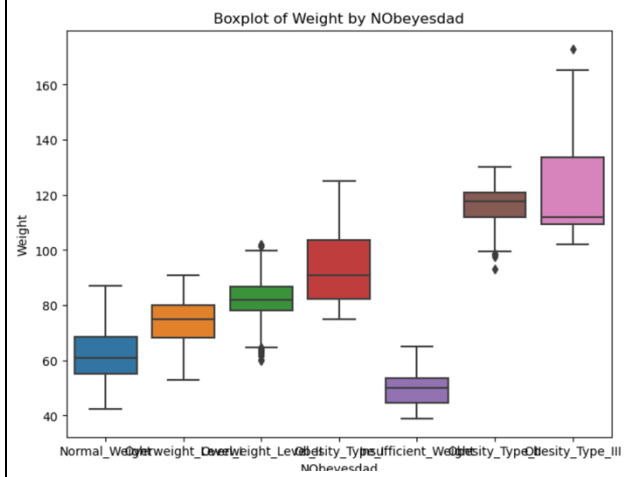


Figure 8

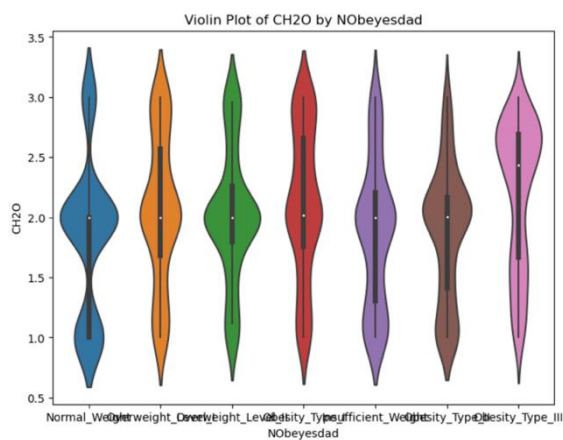


Figure 9

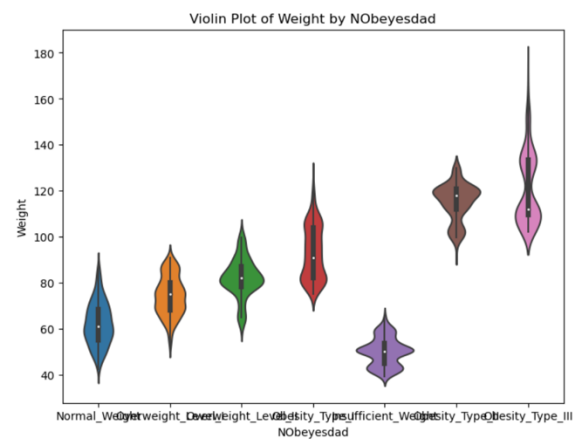
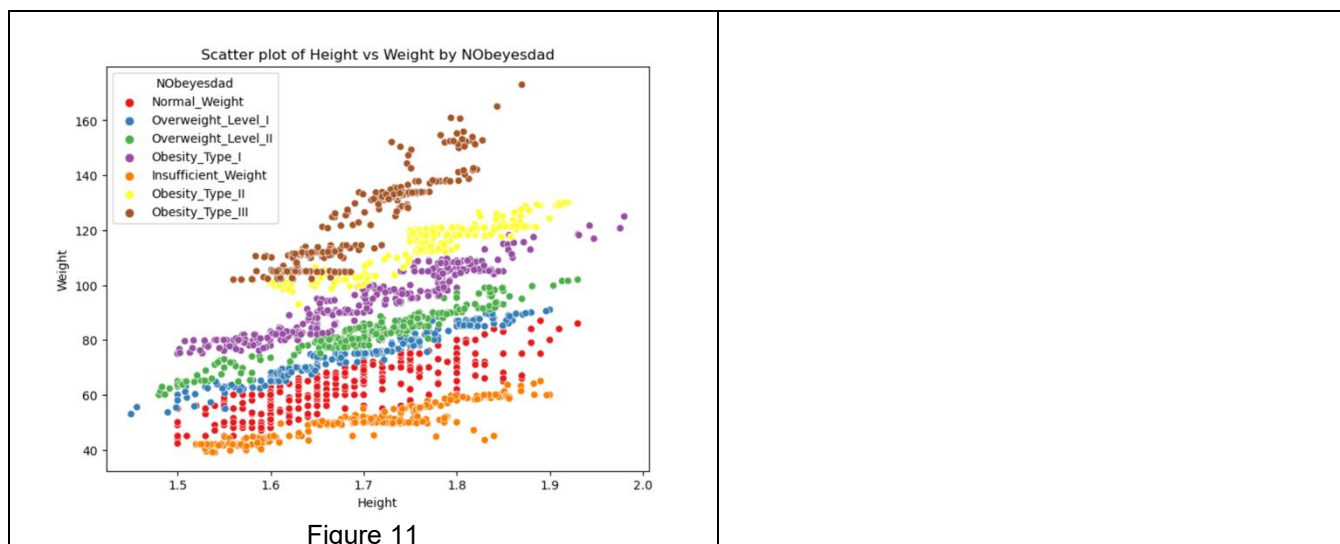


Figure 10



K-NN Algorithm

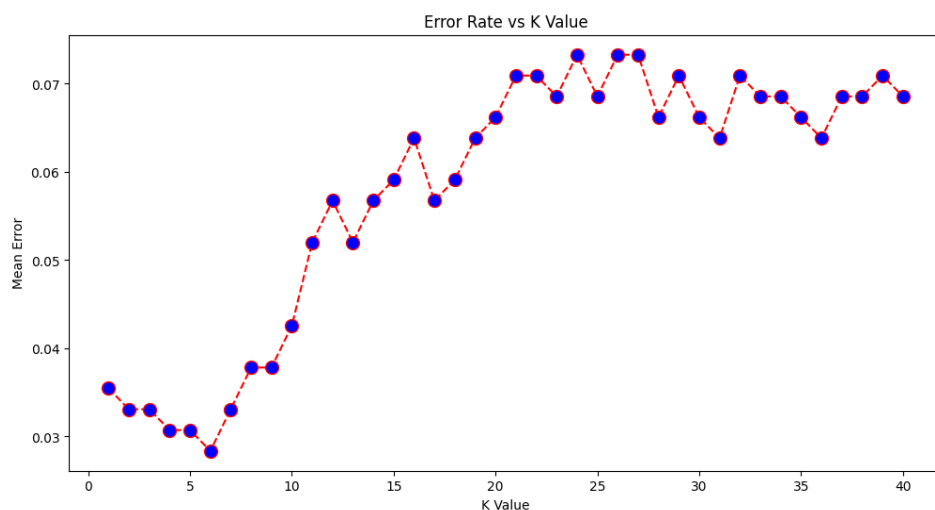
Dataset Preparation and Feature Scaling

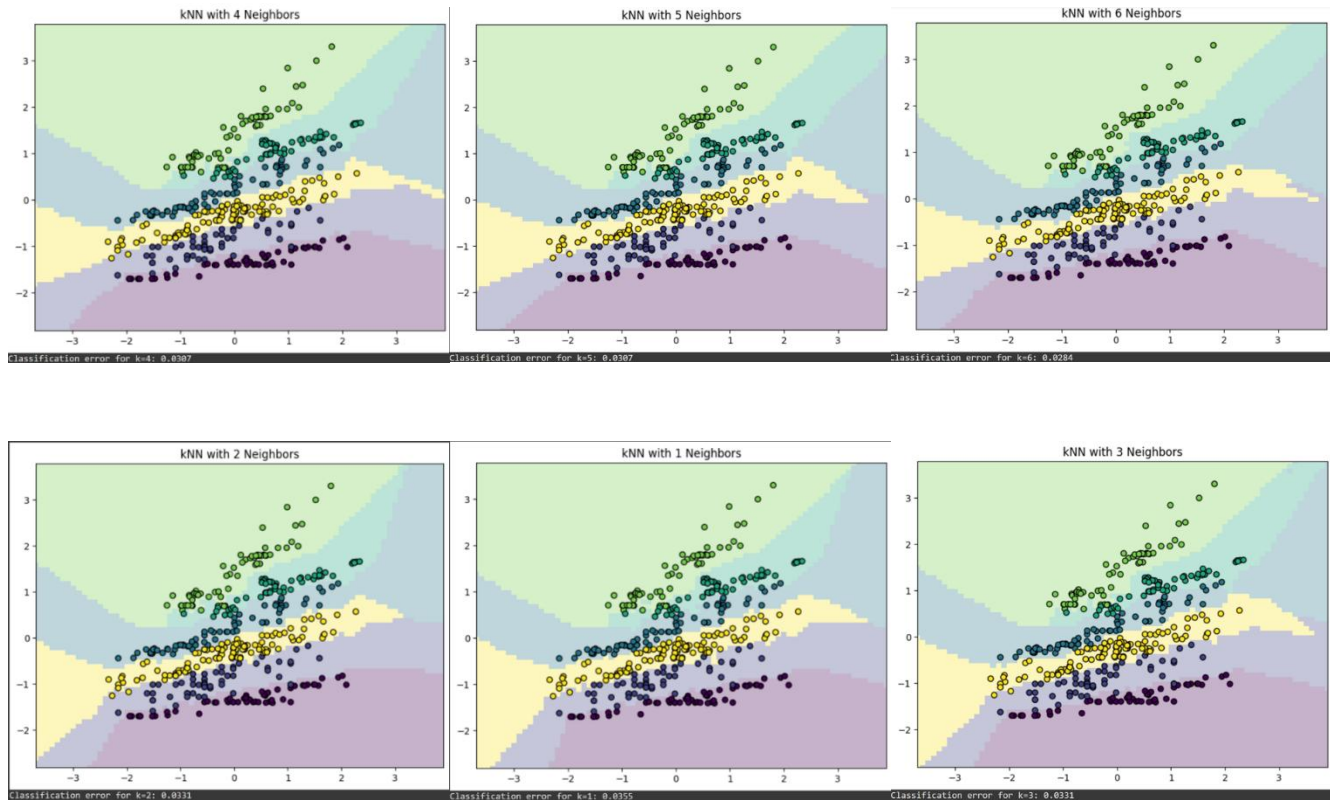
`KNeighborsClassifier()` by default uses Euclidean distance. It is the best distance calculating method in our case, because:

- 1- It effectively captures the geometric closeness between data points
- 2- It's the most common method
- 3- It works best for numeric continuous data such as many features in our Obesity dataset including Age, Height, and Weight
- 4- It's simpler to calculate and interpret
- 5- After scaling using `StandardScaler` ensures fair calculation

Choosing the Right Value of K

By observing the following graphs it is apparent $K=6$ is the best fit.





Model Performance

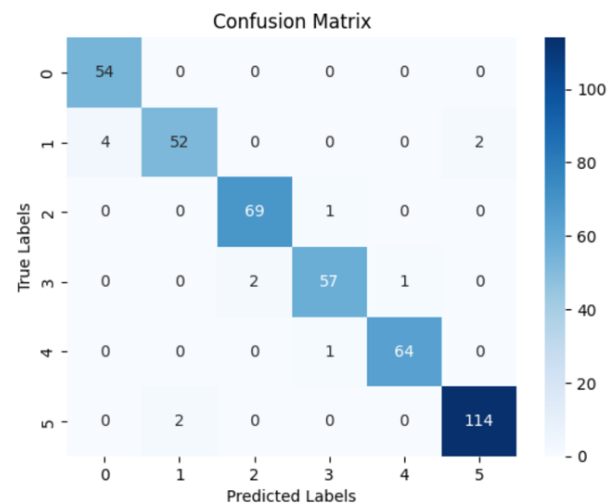
The model achieved an accuracy of 97%, which reflects excellent performance. There are very few misclassifications as observed in the confusion matrix.

Accuracy: 0.9716312056737588

Classification Report:				
	precision	recall	f1-score	support
0	0.93	1.00	0.96	54
1	0.96	0.91	0.94	58
2	0.97	0.99	0.98	70
3	0.97	0.95	0.96	60
4	0.98	0.98	0.98	65
5	0.99	0.98	0.99	116
accuracy			0.97	423
macro avg	0.97	0.97	0.97	423
weighted avg	0.97	0.97	0.97	423

Confusion Matrix:

```
[[ 54  0  0  0  0  0]
 [  4 53  0  0  0  1]
 [  0  0 69  1  0  0]
 [  0  0  2 57  1  0]
 [  0  0  0  1 64  0]
 [  0  2  0  0  0 114]]
```



Cross-Validation

K-Fold was chosen as it provides a reliable estimate of the model's accuracy by using all points for training and testing, with different combinations

10 folds were chosen as that is standard, and the results show relative consistency in the accuracy levels attained.

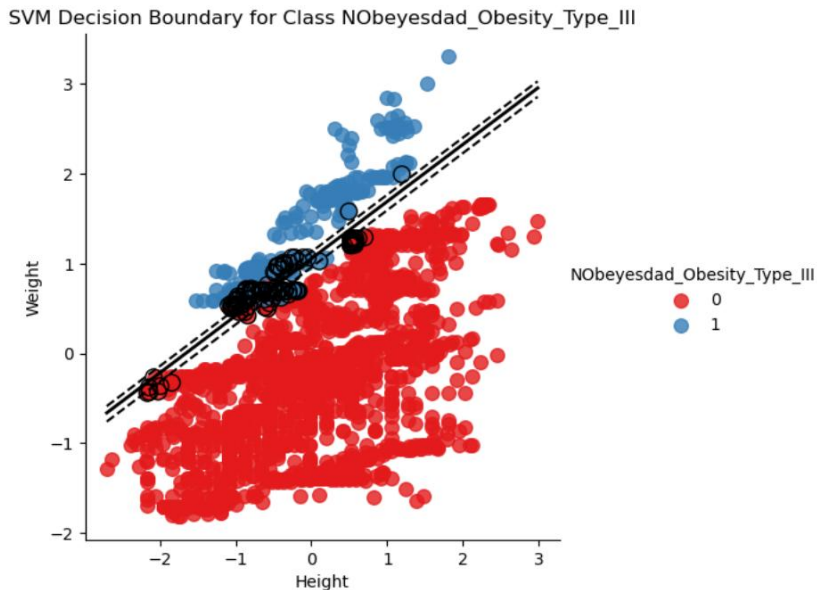
```
Fold Accuracy: 0.9704142011834319
Fold Accuracy: 0.9526627218934911
Fold Accuracy: 0.9822485207100592
Fold Accuracy: 0.9526627218934911
Fold Accuracy: 0.9881656804733728
Fold Accuracy: 0.9763313609467456
Fold Accuracy: 0.9349112426035503
Fold Accuracy: 0.9408284023668639
Fold Accuracy: 0.9345238095238095
Fold Accuracy: 0.9702380952380952
```

SVM Algorithm

Dataset Preparation and Feature Scaling

Since SVM is sensitive to data scaling, StandardScaler() was used on the numerical data for (Age, Weight, and Height). This scaling allows for the data points to line up with the decision boundary when plotted on a graph which is important to the support vectors. Additionally, all categorical columns were encoded although only the height and weight columns were used.

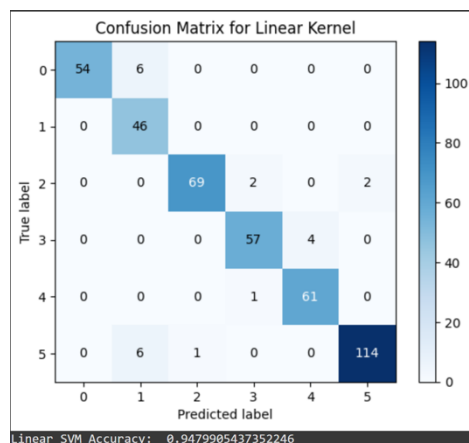
Support vectors



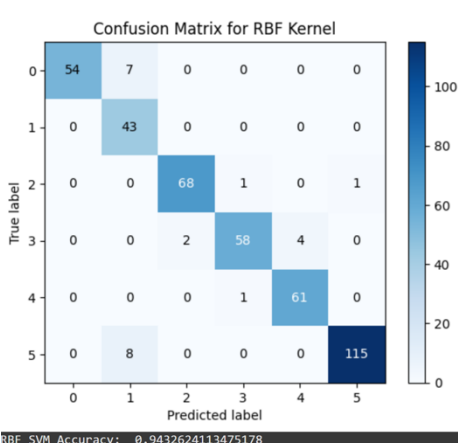
Only the sets of points that maximize the margin can be support vectors as such not all points near the decision boundary are found to be support vectors (This can be noted by looking at the circles which represent the support vectors). By maximizing the margin, the SVM can perform its classification with less chance of error which improves the model's accuracy score.

Kernel Functions

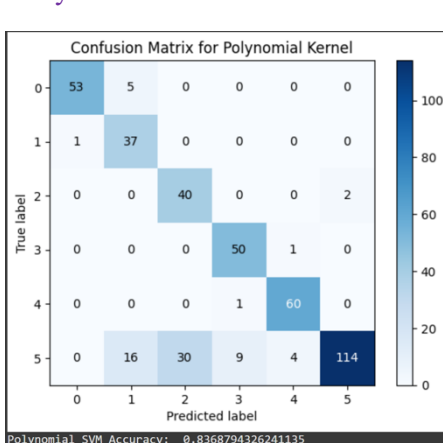
Linear Kernel



RBF Kernel



Polynomial Kernel



While the data isn't separable using one linear line it still can be separated by linear lines for each class which is why the linear kernel performed the best according to its accuracy score. The separation can be done by turning the multiclass problem into a binary class problem where it becomes one class versus the rest. This works directly on the first and last classes. The middle classes will require a projection onto a higher dimension but can still be separated afterwards. This method of splitting the multiclass problem into a binary one is done automatically by the SVM implementation when it detects multiple classes in its output.

Model Performance

```

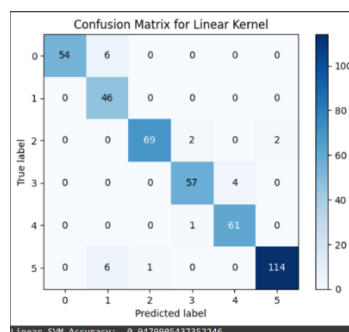
Classification Report:
      precision    recall  f1-score   support

     0       0.90      1.00      0.95        54
     1       1.00      0.79      0.88        58
     2       0.95      0.99      0.97        70
     3       0.93      0.95      0.94        60
     4       0.98      0.94      0.96        65
     5       0.94      0.98      0.96       116

 accuracy          0.95      0.94      0.95      423
 macro avg          0.95      0.94      0.94      423
 weighted avg       0.95      0.95      0.95      423

Confusion Matrix:
[[ 54  0  0  0  0  0]
 [ 4 53  0  0  0  1]
 [ 0  0 69  1  0  0]
 [ 0  0  2 57  1  0]
 [ 0  0  0  1 64  0]
 [ 0  2  0  0  0 114]]

```



The linear kernel's accuracy was the best among the other kernels (approx. 95%). It had less misclassifications (FP and FN) and more correct classifications in total (TP and TN).

Cross-Validation

K-fold cross validation was used on the linear SVM model, and it shows that accuracy levels remain consistent. 10 folds were chosen as that is standard. The standard k-fold method was chosen over the stratified one because the dataset was found to be well balanced meaning there was no need for stratifying.

```

Fold accuracy: 0.9339622641509434
Fold accuracy: 0.9715639810426541
Fold accuracy: 0.9241706161137441
Fold accuracy: 0.933649289099526
Fold accuracy: 0.9478672985781991
Fold accuracy: 0.9478672985781991
Fold accuracy: 0.9289099526066351
Fold accuracy: 0.9715639810426541
Fold accuracy: 0.933649289099526
Fold accuracy: 0.957345971563981

```

Deep Learning/CNN Algorithm

Dataset Preparation and Feature Scaling

In preparation for training the Deep Neural Network (DNN), all numerical features (Age, Height, Weight, etc.) were scaled using StandardScaler to ensure equal contribution of all features and avoid domination of features with higher numeric ranges. Additionally, categorical variables were encoded using one-hot encoding to make them suitable for the neural network. This ensures better convergence and performance of the model by providing a normalized and structured dataset.

Network Architecture Design

A multilayered DNN was used to perform the classification task. It has one input layer that has the shape of 16 (number of features) and a dense output layer with the number of classes as the number of neurons (6) since that is the most suitable with such multiclass classification problem. It also has a variable number of hidden layers (either two or three) to evaluate the performance with each option. The number of neurons is also variable (128,256,512) with each layer being half the size of the previous one. This is to avoid overfitting and ensure real representation of the data. Since the problem is somewhat complex having 6 nonlinearly separable classes, we needed to experiment with different values for the parameters to reach the best outcome.

Upon training and testing the model, it was found that the more neurons and number of layers, and the lower batch size, the better the accuracy was. We found the optimum parameters to be {512 neurons, 32 batch size, 3 hidden layers}

Input Layer: 16 neurons (equal to number of features).

Hidden Layers: 3 layers with 512 → 256 → 128 neurons.

Output Layer: 6 neurons (one for each obesity class).

Activation Functions:

Hidden Layers: ReLU (best for non-linear problems)

Output Layer: Softmax (for multiclass classification)

Batch Size: 32 (best balance between stability and speed)

Input Layer: Accepts raw feature data.

Hidden Layers: Detect patterns and relationships.

Output Layer: Provides final class probabilities.

Activation Functions

Since it is a multiclass problem, softmax activation function works best in classification of the output layer. Relu was used for the hidden layers due to its ability to deal with their complexity. With these choices we got the best performance of our model on the testing and predicting the true classes. Batch Size: Larger batch sizes speed up training but may generalize poorly. Smaller batch sizes provide better generalization but increase training time.

Hyperparameter Tuning

A higher learning rate can speed up the learning process, but it might skip the optimal solution. A lower learning rate takes longer to learn, but it's more likely to find the best solution. Finding the right rate is key for good performance.

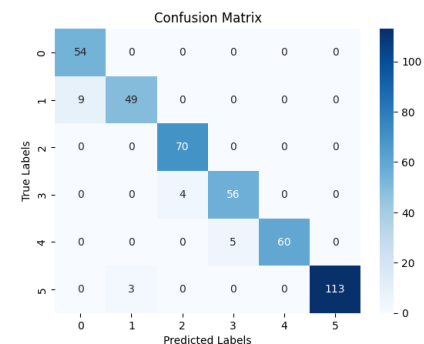
CNN

Inapplicable

Model Performance

```
Precision: 0.9533
Recall: 0.9504
F1-Score: 0.9503
```

			precision	recall	f1-score	support
	0	0.86	1.00	0.92	54	
	1	0.94	0.84	0.89	58	
	2	0.95	1.00	0.97	70	
	3	0.92	0.93	0.93	60	
	4	1.00	0.92	0.96	65	
	5	1.00	0.97	0.99	116	
accuracy			0.95		423	
macro avg	0.94	0.95	0.94	423		
weighted avg	0.95	0.95	0.95	423		



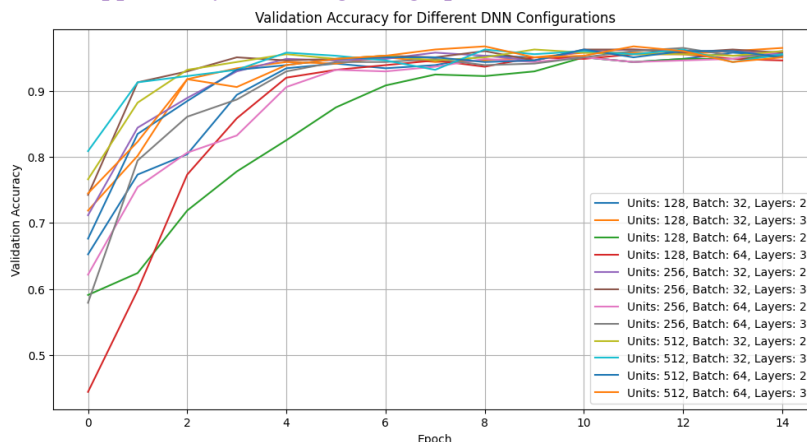
The accuracy of the model is 95%, indicating strong overall performance. There is high precision and recall which reflects a good balance between correctly identifying positive cases and minimizing false positives. While the model performs well, precision for class 1 (0.86) could be improved to reduce false positives.

Validation

K-fold cross-validation can be used with DNNs, but it's less common due to the high computational cost. For DNNs, performance is typically tracked through epochs. Instead of re-training the model multiple times like in k-fold, the model's performance is evaluated after each epoch. This approach is computationally more efficient while still providing insight into the model's ability to generalize.

The model was trained over 15 epochs with varying numbers of neurons, hidden layers, and batch sizes. This shows how accuracy varies with different hyperparameters, achieving cross-validation and ensuring the best hyperparameters were identified.

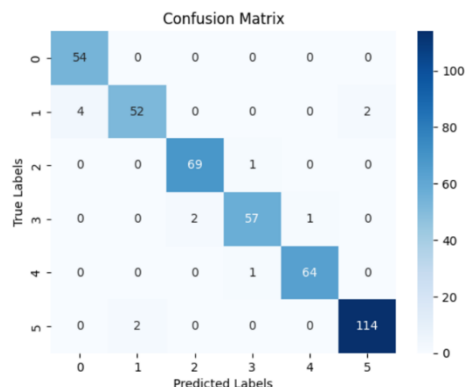
This is apparent by observing this graph which shows validation accuracy vs epoch:



Comparison between KNN, SVM and deep NN/CNN.

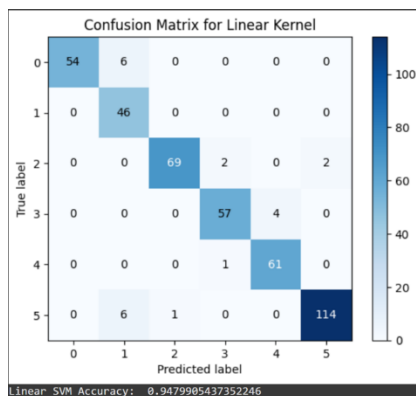
Confusion matrix for:

1- KNN:



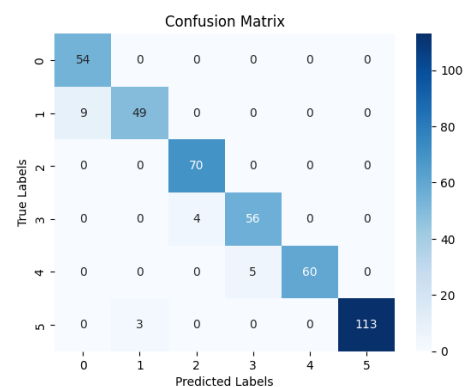
With accuracy = 97%

2- SVM:



With accuracy = 95%

3- DNN:



With accuracy = 95

Comparison:

KNN has the highest overall accuracy

Class	KNN Recall	DNN Recall	SVM Recall	Best Recall Model
0	1.00	1.00	1.00	All
1	0.91	0.84	0.79	KNN
2	0.99	1.00	0.99	DNN
3	0.95	0.93	0.95	KNN+SVM
4	0.98	0.92	0.94	KNN
5	0.98	0.97	0.98	KNN+SVM

By comparing the recall values for each model, it is apparent that KNN works best for most classes. For class #2 DNN outperforms KNN but only by 0.01, therefore, KNN is considered the best fit.

Conclusion

Students majoring in computer science and engineering are concerned with learning how to analyze any kind of data to improve a wide variety of fields, including the health sector. Bioengineering students can develop technologies that will help improve people's health; and understanding which factors cause issues like obesity is the first step in solving the problem.

Work Distribution

	REEMA	LENA	SARAH	FATMAH
Data Prep + Visualization	8%	8%	76%	8%
KNN	76%	8%	8%	8%
SVM	8%	76%	8%	8%
DNN	8%	8%	8%	76%
Report Template	10%	10%	70%	10%

Each team member contributed an equal amount of work into the project.